

OPJ Girls College F-8/2, Islamabad
FIRST TERM EXAMINATION 2026
CHEMISTRY SSC-I

Marks: 39

Time allowed: 2 hr 15 minutes

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the provided sheet.

SECTION-B (Marks 27)

Note: Attempt all questions.

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1	Differentiate between concentrated and dilute solutions.	3	OR	Write empirical formula of Nicotine ($C_{10}H_{14}N_2$) AND Caffeine ($C_8H_{10}N_4O_2$)	3
2	How a colloidal dispersion is distinguished from true solution?	3	OR	What is the effect of temperature on solubility.	3
3	Why is an atom electrically neutral?	3	OR	Define oxidation number. What is the common oxidation state of group 1A and VIA elements in binary compounds.	3
4	Arrange in increasing ionization energy: lithium, sodium, potassium (with reason). $Li=3$, $Na=11$, $K=19$	1+1+1	OR	Arrange in decreasing shielding effect: chlorine, bromine, iodine (with reason). $Cl=17$, $Br=35$, $I=53$	3
5	Why does transition elements show variable oxidation state.	3	OR	Represent the formation of cation in (a) Al $Al=13$ (b) Ca $Ca=20$	1.5+ 1.5
6	Convert into ionic equation: $Mg_s + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + H_{2(g)}$	3	OR	Write a balanced equation for burning of hydrogen to produce water reaction.	1+2
7	Derive formula of aluminum sulfate.		OR	Find oxidation state of nitrogen in HNO_3 and NO_2 ($H=1, O=8$)	1.5 +1.5
8.	Calculate the enthalpy of the reaction between hydrogen and chlorine to form hydrogen chloride from the following bond energy data: Bond energy of $H-H = 436 \text{ kJ mol}^{-1}$ Bond energy of $Cl-Cl = 243 \text{ kJ mol}^{-1}$ Bond energy of $H-Cl = 432 \text{ kJ mol}^{-1}$	3	OR	What is the role of a catalyst in a chemical reaction?	3
9	Differentiate between forward and reverse reactions with example.	1.5 +1.5	OR	How does temperature effect cobalt chloride at equilibrium.	3

SECTION- C (Marks 12)

Note: Attempt all questions. Marks of each question are given along with each question.

1.	Identify the oxidizing agents and reducing agents in the following reactions: 1. $H_2S + Cl_2 \rightarrow 2HCl + S$ 2. $2FeCl_2 + Cl_2 \rightarrow 2FeCl_3$ 3. $Mg + 2HCl \rightarrow MgCl_2 + H_2$	2+2 +2	OR	Describe the formation of covalent bond in following compounds a. N_2 b. NH_3 c. O_2	2+2 +2
2.	Define exothermic and endothermic reactions. Draw a labeled reaction pathway diagram for an exothermic and endothermic reaction.	2+4	OR	Calculate the mass of 3×10^{10} molecules of Oxygen gas. $N=14$ $O=16$	6

formula

NA

4.98×10^{12}